

Test and Reliability Exercise Sheet 4

Exercise 1

Assume a fault list $F = \{f_0, \dots, f_{10}\}$ and the following test set, which was generated in the given order, i.e. t_0 was generated first and t_7 was generated last. For each test t_i all faults are given which are detected by this test.

- $t_0 = \{f_0\}$
- $t_1 = \{f_0, f_1, f_2\}$
- $t_2 = \{f_2, f_3\}$
- $t_3 = \{f_4, f_7\}$
- $t_4 = \{f_5, f_6\}$
- $t_5 = \{f_4, f_7, f_8\}$
- $t_6 = \{f_9\}$
- $t_7 = \{f_8, f_9, f_{10}\}$

Perform the *Forward-Looking Reverse Order Fault Simulation* for the given test set. Identify for each fault f_j the test t_i which first detects f_j . Then, simulate the test set in reverse order and skip each test t_i which is not the first test to detect any yet undetected fault.

Exercise 2

- a) State for the following set of 2-input gates the controlling values as well as the non-controlling ones: $\{AND, NAND, OR, NOR, XOR, XNOR\}$
- b) Reconsider the example circuit for static test pattern relaxation from the lecture as given in Fig. 1. In the lecture we obtained the test pattern $t_g = 1X11$ (by utilizing a greedy approach) and $t_o = 11XX$ (as optimal solution in number of unspecified inputs) for the stuck-at-fault $s@0$ at the input A (see Fig. 1). Find another stuck-at-fault in the circuit for which a (relaxed) test pattern can be statically compacted with t_g , but not with t_o .

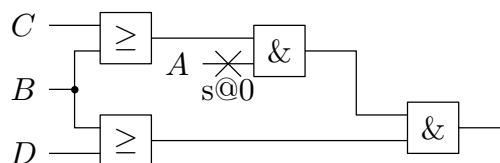


Figure 1: Example circuit

Exercise 3

- Perform a cone-of-influence (COI) reduction on the circuit in Fig. 2 with a stuck at 0 fault at line x.
- Build a miter circuit with the relevant part of the circuit.

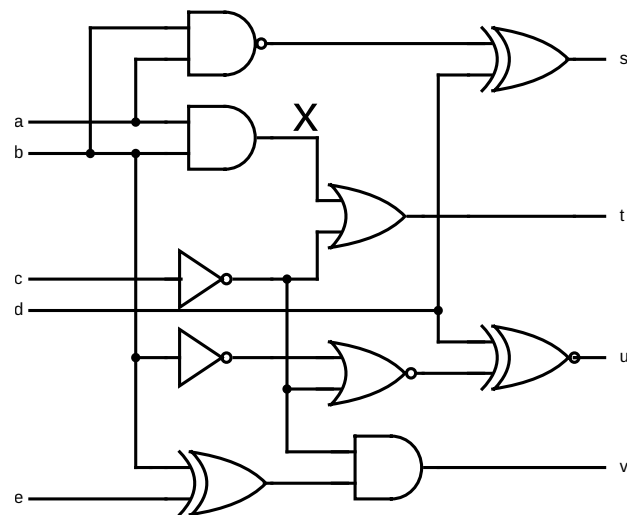


Figure 2: Cone-of-Influence reduction circuit

Exercise 4

- a) Perform a Tseitin transformation on the circuit from the lecture as given in Fig. 3.
- b) Perform the SAT-Solving algorithm (chapter 4, slides 35ff) from the lecture on the SAT-instance you obtained from part a) including the unit clause “set BD to 1”. Whenever you reach a conflict, go back to the last decision you have made. Additionally, use the following decision heuristic:
 - When you have to make a decision, choose variables close to the inputs of the original circuit first. If there are still multiple possibilities, choose an arbitrary one.
 - When you have chosen a variable, assign the logical value 0 first to this variable.
- c) Compare the circuit with your solution of exercise 3b.

Deadline: Discussion on Thursday, June 25th, 2026

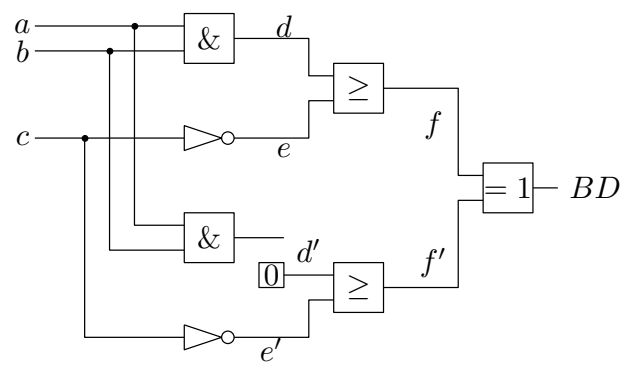


Figure 3: Miter circuit